

Lightweight MobileViT-Graph Network for Topological Rural Road Extraction

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- NetworkX Planar Graph $G=(V,E)$ & Betweenness Centrality
- Dependency-Free ONNX Engine (38.5 ms CPU, 0.82 MB size)

3. Encoder Deep-Dive: Layer-by-Layer Mechanics

3.1 Stem: Factorized 1D Strip Convolutions (1x3 -> 3x1)

Standard convolutional stems utilize square 3x3 filters (9 weights per channel pair) that waste parameters learning isotropic shapes. Our stem factorizes downsampling into a horizontal filter (1x3) followed by a vertical filter (3x1):

Mathematical Formulation:

$$X_h = \text{SiLU}(\text{BatchNorm}(\text{Conv}_{\{1 \times 3\}}(X, \text{stride}=(1, 2), \text{padding}=(0, 1))))$$

$$X_{\text{stem}} = \text{SiLU}(\text{BatchNorm}(\text{Conv}_{\{3 \times 1\}}(X_h, \text{stride}=(2, 1), \text{padding}=(1, 0))))$$

Analogy — Paint Roller vs. Square Stamp: A square stamp stamps indiscriminately in all directions. A horizontal and vertical paint roller scans along orthogonal axes, matching the linear geometry of roads and ignoring square farm plots. It reduces convolutional parameters by exactly **33.3%**.

3.2 Stage 1: MobileNetV2 Inverted Residual Bottlenecks

Stage 1 processes features through MobileNetV2 inverted residual bottlenecks with expansion ratio $e=2$. It applies a 1x1 pointwise expansion (32 -> 64 channels), a 3x3 depthwise separable convolution at stride 2 (128 -> 64 resolution), and a linear 1x1 projection without non-linear activation to prevent destruction of manifold topology.

3.3 Novelty: Zero-Parameter Channel Shift

Before transformer blocks, we expand the spatial receptive field at **zero parameter and zero FLOP cost**. Given intermediate tensor Z (C channels), a fraction $\gamma = 0.25$ (16 channels) is partitioned into four equal subsets and physically displaced by $s = 2$ pixels:

- **Upward Group (4 chan):** Shifted up by 2 pixels; vacated bottom row is zero-padded.
- **Downward Group (4 chan):** Shifted down by 2 pixels; vacated top row is zero-padded.
- **Leftward Group (4 chan):** Shifted left by 2 pixels; vacated right column is zero-padded.
- **Rightward Group (4 chan):** Shifted right by 2 pixels; vacated left column is zero-padded.
- **Identity Group (48 chan):** Passed through completely unchanged.

Analogy & Physical Road Match: Like sending four scouts 2 steps in cardinal directions while the commander stays centered. At 4x downsampling, a 2-pixel displacement corresponds to **± 8 pixels in original satellite space**. In 0.5m–1.0m imagery, 8 pixels is roughly 4 to 8 meters—the **exact physical width of an Indian rural road!**

3.4 Separable Linear Self-Attention ($O(N \cdot d)$ vs $O(N^2)$)

Standard Vision Transformers compute scaled dot-product attention $\text{Attention}(Q, K, V) = \text{Softmax}(Q \cdot K^T / \sqrt{d}) \cdot V$. For $N = 1,024$ tokens, the $N \times N$ affinity matrix requires over 1,000,000 operations per head per tile, causing memory exhaustion on edge devices.

MobileViT v2 replaces token-to-token affinity with **Global Context Aggregation**:

1. Linear projection maps input tokens to Query Q (dim d), **Scalar Key K (dim 1)**, and Value V (dim d).
2. Softmax is evaluated along the token dimension: $\alpha = \text{Softmax}(K, \text{dim}=N)$ in $R^N(B \times N \times 1)$.
3. A single Global Context Vector is computed: $c = \sum_{n=1}^N (\alpha_n \cdot V_n)$ in $R^d(B \times 1 \times d)$.
4. Token queries are modulated by the global vector: $\text{Output} = \text{LinearProj}(\text{ReLU}(Q) \blacksquare c)$.

Analogy — Loudspeaker vs. Private Whispers: Standard attention is like 1,000 people having private 1-on-1 conversations (1,000,000 interactions). Linear attention is like one speaker summarizing everyone's notes and broadcasting a single announcement over a loudspeaker (1,000 interactions). Complexity drops from $O(N^2)$ to **$O(N \cdot d)$** , running 1,000x faster!

3.5 Deep Bottleneck Layer

At the lowest resolution (16x16 grid, 128 channels), three stacked MobileViT v2 blocks with embedding dimension $d=192$ process latent features. Each token corresponds to a 16x16 meter patch of physical terrain, enabling the model to bridge vast 100-meter canopy occlusions.

4. Attention Gates: Intelligent Skip Filtering

In standard U-Net, skip connections copy raw encoder features directly to the decoder. In rural remote sensing, early encoder features are flooded with non-road clutter: plow furrows, crop boundary berms, and tin roofs.

We deploy Additive Attention Gates across all three skip connections:

$$q_att = \text{SiLU}(W_g * g + W_s * s + b_g + b_s)$$

$$\alpha = \text{Sigmoid}(\psi * q_att) \text{ in } [0.0, 1.0]$$

$$s_gated = s \cdot \alpha$$

The Airport Security Analogy: The deep decoder acts as a supervisor: 'I know roughly where the road corridor is' (Gating signal g). The encoder skip arrives with raw sharp edges (Skip signal s). The Attention Gate compares them: pixels on genuine roads get $\alpha \approx 1.0$ (pass through), while background farm fields and trees get $\alpha \approx 0.0$ (blocked). The decoder receives clean road boundaries with zero jungle clutter.

5. Decoder Deep-Dive: Multi-Scale Reconstruction

The decoder reconstructs the spatial probability map through three expanding stages (16 -> 32 -> 64 -> 128 -> 256):

- **Bilinear Upsampling 2x:** Doubles spatial resolution smoothly without checkerboard deconvolution artifacts.
- **Skip Fusion:** Concatenates upsampled tensor with the gated skip tensor.
- **Strip Convolution Fusion:** Fuses features using 1x3 -> 3x1 filters to enforce linear road geometry.
- **Final Segmentation Head:** A 1x1 convolution maps 32 channels to a single-channel road probability map with Sigmoid activation.

6. Differentiable Topological Supervision (clDice Loss Engine)

Binary Cross-Entropy (BCE) measures volumetric pixel overlap. A thin 3-pixel road broken by a 10-pixel canopy shadow represents only 30 mispredicted pixels (0.045% of scene area). The BCE loss penalty for this gap is negligible, allowing traditional networks to leave roads permanently fragmented. We overcome this by supervising on morphological skeletons using the differentiable **Centerline-Dice (clDice)** loss engine.

6.1 Soft Morphological Skeletonization

To backpropagate gradients through morphological thinning, we implement soft morphological primitives using min/max pooling:

$$\text{soft_erode}(I) = \min(-\text{MaxPool}_{\{3 \times 1\}}(-I), -\text{MaxPool}_{\{1 \times 3\}}(-I))$$

$$\text{soft_dilate}(I) = \text{MaxPool}_{\{3 \times 3\}}(I)$$

$$\text{soft_open}(I) = \text{soft_dilate}(\text{soft_erode}(I))$$

Iterative soft skeletonization is evaluated recursively over $K=10$ iterations: $S_{\{k+1\}}(I) = S_k(I) + \text{ReLU}(\Delta_{\{k+1\}} - S_k(I) \cdot \Delta_{\{k+1\}})$. Every operation is continuous and fully differentiable in PyTorch Autograd.

6.2 Soft Centerline-Dice Formulation

Given predicted soft skeleton $S(P)$ and ground truth skeleton $S(Y)$:

- Topological Precision: $T_prec = (|S(P) \cap Y| + \epsilon) / (|S(P)| + \epsilon)$
- Topological Sensitivity: $T_sens = (|P \cap S(Y)| + \epsilon) / (|S(Y)| + \epsilon)$
- clDice Loss: $L_clDice = 1.0 - (2 \cdot T_prec \cdot T_sens) / (T_prec + T_sens + \epsilon)$

Whenever a road prediction drops to zero under a tree canopy while the ground-truth skeleton crosses through, the numerator $|P \cap S(Y)|$ drops severely, generating massive backpropagated gradients directly at the occluded pixels.

6.3 Dynamic Alpha Decay & Collapse Gating

In early training, pure clDice causes 'topological collapse': the network blanket-predicts road everywhere to minimize loss. We inoculate the model via three mechanisms: (1) Clamping BCE positive class weight to $w_pos = 2.0$, (2) Implementing power-law alpha decay $\alpha(e) = 0.40 + 0.10 \cdot (1 - \min(1, e/40))^{0.5}$ to anchor learning with BCE early on, and (3) Enforcing hard-gated canary checkpoint validation that automatically aborts any epoch with positive road pixel fraction > 15%.

7. Geometric Post-Processing & Spatial Graph Vectorization

Once probability maps are output by the model, a four-step post-processing pipeline converts rasters into navigable planar graphs:

7.1 Hysteresis Thresholding (Faint Road Recovery)

Under tree canopies, probability predictions attenuate to faint levels (p in $[0.15, 0.35]$). We apply dual thresholds: High Threshold $T_{high} = 0.35$ (seed road pixels) and Low Threshold $T_{low} = 0.12$ (shadowed continuation pixels). Faint road pixels ($p \geq 0.12$) are preserved if and only if they form a contiguous path connecting to a high-confidence seed ($p \geq 0.35$). Isolated false positives in farm fields are discarded.

7.2 Skeleton Endpoint & Outward Tangent Vector Extraction

Binary masks are thinned to a 1-pixel skeleton S and convolved with an 8-neighborhood degree kernel $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 10 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. Pixels with value 11 have exactly one neighbor, mathematically identifying them as **dead-end endpoints**. For each dead-end e_i , we compute its outward unit tangent velocity vector v_i by calculating the spatial centroid of its backward 5-pixel neighborhood.

7.3 Multi-Strategy Canopy Gap Bridging

- **Strategy A (Collinear Facing Endpoints):** For candidate pairs (e_i, e_j) within Euclidean distance $d \leq 220$ pixels (~110m on ground), we check if their outward tangent vectors face each other within a 65° angular cone ($v_i \cdot u_{ij} \geq \cos(65^\circ)$). If validated, a vector bridge is drawn.
- **Strategy B (T-Junction Completion):** Dead-end endpoints are projected forward along v_i toward perpendicular trunk skeletons within 220 pixels, reconnecting obscured rural T-junctions.

7.4 Vector Planar Graph & Criticality Analysis

Skeletons are vectorized into an undirected NetworkX graph $G = (V, E)$ where intersections form Junction nodes (degree ≥ 3) and road segments form Edges with geodesic lengths in meters.

- **Betweenness Centrality $C_B(v)$:** Evaluates the fraction of all network shortest paths passing through node v . Peak nodes are designated 'Gatekeeper Nodes'—critical arterial bridges whose failure isolates entire villages.
- **Resilience Index (R):** Simulates targeted disaster ablation: $R = \text{Efficiency}(G \text{ without top } 10\% \text{ gatekeepers}) / \text{Efficiency}(G_{\text{baseline}})$. An index $R < 0.50$ alerts PMGSY authorities of severe single-point failure vulnerability.

8. Edge Deployment Architecture (ONNX Runtime)

To operate in remote areas without cloud connectivity, PyTorch weights are compiled to an ONNX graph via `backend/scripts/export_onnx.py`. Sigmoid activation is baked directly into the graph, yielding probability rasters directly with dynamic tile dimensions (H, W). The production runtime script `scripts/predict_onnx.py` requires only `onnxruntime`, `opencv`, and `numpy` (under 80 MB container footprint). It evaluates a tile in **38.5 ms on a standard laptop CPU** with numerical parity verified to $|y_{\text{PyTorch}} - y_{\text{ONNX}}| \leq 2.14e-6$.

9. Empirical Benchmarks & Ablation Verification

Architecture	Parameters	Payload Size	IoU (%)	cIDice (Connectivity)	APLS (Navigability)	CPU Latency
Baseline U-Net	31.04 M	118.4 MB	58.42%	52.14%	44.20%	142.0 ms
DeepLabv3+	54.70 M	208.9 MB	61.20%	56.32%	48.60%	285.0 ms
MobileViT_v2 (BCE only)	1.60 M	1.83 MB	63.80%	57.40%	49.80%	38.5 ms
Proposed MobileViT-Graph	1.60 M	0.82 MB (ONNX)	74.62%	81.62%	76.80%	38.5 ms

Ablation Stage	Configuration	cIDice (%)	APLS (%)	Modular Gain Source
Stage 1	U-Net Baseline (BCE)	52.14%	44.20%	Baseline control reference
Stage 2	MobileViT_v2 (BCE only)	57.40%	49.80%	94.8% parameter compression
Stage 3	MobileViT_v2 + cIDice	78.54%	72.40%	+21.1% cIDice via topological loss
Stage 4	Full Proposed Framework	81.62%	76.80%	+32.6% APLS via gap healing & weak labels

10. Viva & HOD Presentation Q&A; Guide

Q1: Why use MobileViT instead of standard U-Net or ResNet?

U-Net relies solely on local 3x3 convolutions. When a rural dirt road passes under a tree canopy or dense forest, local pixel information is blacked out, and local convolutions assume the road ended. MobileViT v2 introduces linear self-attention, giving the model a global receptive field to 'see' that a road entering the forest matches the trajectory of a road exiting on the other side, successfully bridging the occlusion.

Q2: What makes your attention mechanism 'Linear' ($O(N \cdot d)$) instead of standard Transformer attention ($O(N^2)$)?

Standard attention computes a token-to-token dot-product affinity matrix of size $N \times N$, which explodes quadratically. MobileViT v2 projects the Key into a scalar (dimension 1) instead of a vector. Applying Softmax across the token dimension produces scalar weights that combine all Value tokens into a single Global Context Vector of size $1 \times d$. Each Query token is then modulated by this single global vector. Computational cost scales as $O(N \cdot d)$, which is linear with respect to pixel count.

Q3: Why did you implement 'Strip Convolutions' instead of standard square convolutions?

Rural roads are elongated tubular corridors (3–8 pixels wide) running horizontally, vertically, or diagonally. Standard 3x3 square kernels waste capacity learning isotropic square textures (such as houses or crop fields). Strip convolutions factorize the operation into a 1x3 horizontal filter followed by a 3x1 vertical filter. This injects an inductive prior that acts as a directional scanner for roads, while saving 33.3% of parameters.

Q4: Explain the 'Zero-Parameter Channel Shift'. How does it expand the receptive field for free?

Before transformer attention, we take 25% of feature channels and physically displace them in memory by 2 pixels along the four cardinal directions ($\uparrow, \downarrow, \leftarrow, \rightarrow$), zero-padding edges. Because this is a simple tensor memory slice ($x[:, :, 2:, :]$), it requires zero learned weights and zero FLOPs. At 4x downsampling, a 2-pixel shift equals an 8-pixel reach in original image space—which exactly matches the physical width of a rural dirt road.

Q5: What is the purpose of Attention Gates on skip connections?

Standard U-Net skip connections blindly transfer all high-resolution features from the encoder to the decoder. In rural satellite scenes, early encoder features are full of noise—agricultural furrows, tin roofs, and forest textures. The Attention Gate uses the deeper decoder feature map (which knows the coarse location of the road) as a gating signal to compute a spatial attention mask α in $[0, 1]$. This suppresses background noise while letting true road edges pass through cleanly.

Q6: Why can't we just train with standard Binary Cross-Entropy (BCE) loss?

Rural roads represent less than 3% of total image pixels. If a 3-pixel road is broken by a 10-pixel canopy shadow, mispredicting those 30 pixels accounts for only 0.045% of the total loss. BCE is mathematically blind to connectivity. cDice solves this by computing overlap on the morphological skeleton (centerline) of the road. If a continuous centerline crosses an empty prediction gap, cDice applies a massive gradient penalty directly at the break.

Q7: What is 'Topological Collapse' and how did you prevent it?

Early in training, the network does not yet know where roads are located. If supervised purely with cDice, the model can maximize topological sensitivity by predicting road probabilities everywhere (a completely white image). We prevented this through three safeguards: (1) clamping positive BCE weight to $w_{\text{pos}} = 2.0$, (2) using a power-law alpha decay schedule that anchors the network with BCE early, and (3) hard-gating checkpoints so that any epoch producing more than 15% positive road pixels is automatically discarded.

Q8: How does your tangent-guided gap healing algorithm bridge canopy gaps?

We thin the predicted road mask to a 1-pixel skeleton and convolve it with an 8-neighborhood degree kernel to identify dead-end endpoints (pixels with only 1 neighbor). For each dead-end, we compute its outward unit tangent vector by looking backward 5 pixels along the skeleton. If another endpoint is within 220 pixels and points toward it within a 65° collinear cone, we draw a vector bridge connecting them.

Q9: What is the significance of the Betweenness Centrality analysis on the extracted graph?

Once the road mask is converted into a NetworkX planar graph, Betweenness Centrality calculates which nodes lie on the highest number of shortest paths. In rural networks, nodes with peak centrality are 'Gatekeeper Arterials'—such as single-access bridges connecting remote villages. If that node is severed during a flood or landslide, the network efficiency collapses. This provides actionable intelligence for PMGSY disaster planning.

Q10: How does your model run without PyTorch on an edge device?

We compiled the trained PyTorch checkpoint into an optimized ONNX computation graph with Sigmoid activation baked in and dynamic dimensions enabled. The production inference script (`predict_onnx.py`) requires only `onnxruntime`, `opencv`, and `numpy`. The container footprint is under 80 MB, and the model evaluates a tile in 38.5 ms on a standard laptop CPU without needing a GPU or CUDA.